

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	01 October 2026
Team ID	SWTID-2026-8331
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

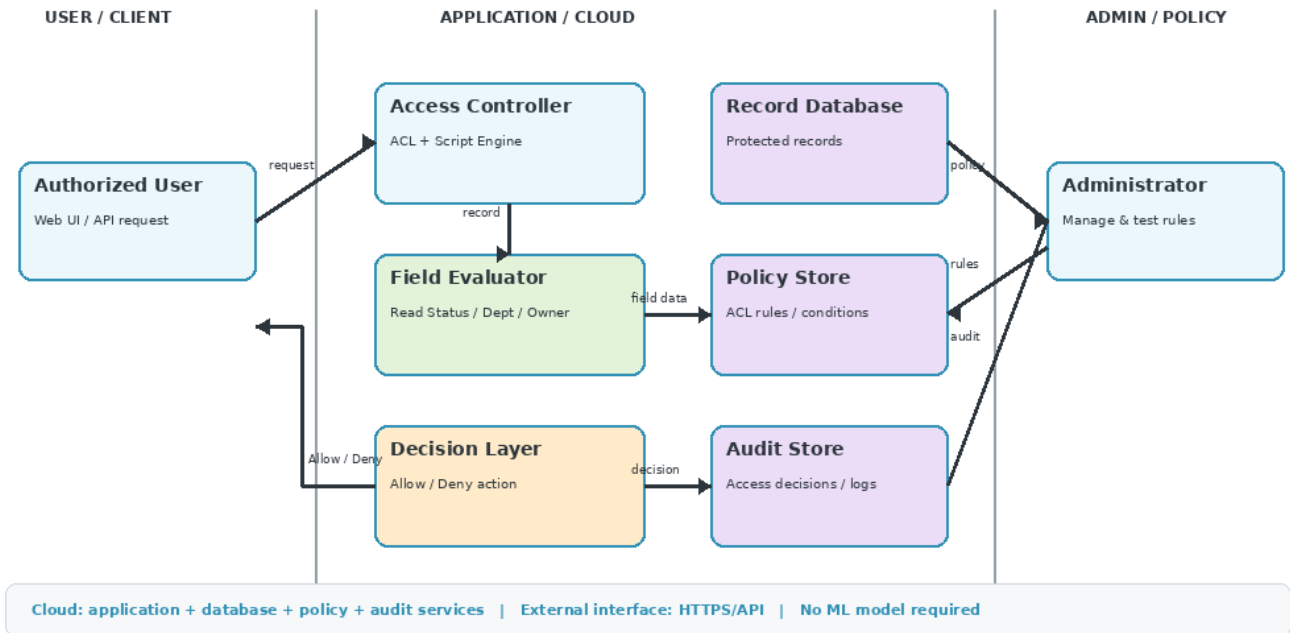
Technical Architecture:

The deliverable includes the project architecture and technology components required to evaluate record access from field values and enforce the resulting ACL decision.

Example: Script-Controlled ACL access flow

Script-Controlled ACL — Technical Architecture

Field-value-driven record access control



Guidelines:

- Include all application processes / technology blocks: request handling, field evaluation, ACL/script evaluation, decision and audit.
- Provide infrastructural demarcation: application and data services are designed for cloud deployment.
- Indicate external interfaces: HTTPS/API access between the client, application and administrative policy interface.
- Indicate data storage components: protected record database, ACL policy store and audit log store.
- Indicate machine-learning interface only if applicable; this ACL solution does not require an ML model for its core access decision.

Table-1 : Components & Technologies:

S.No	Component	Description / Technology
1.	User Interface	How user interacts with application — Web UI / REST API client. Technology: HTML, CSS, JavaScript.
2.	Application Logic-1	Request handling and authentication. Technology: Python / FastAPI.
3.	Application Logic-2	Field-value evaluation and ACL script execution. Technology: Python rule engine / validation logic.
4.	Application Logic-3	Access decision and enforcement. Technology: Python service logic / middleware.
5.	Database	Protected record data and field values. Technology: PostgreSQL / relational database.
6.	Cloud Database	Cloud-hosted database service for records, policies and audit data. Technology: managed PostgreSQL.
7.	File Storage	Optional storage for configuration exports, audit reports or deployment artifacts. Technology: cloud object storage / local filesystem.
8.	External API-1	Purpose: secure client-to-application communication. Technology: HTTPS / REST API.
9.	External API-2	Purpose: optional identity/authentication integration. Technology: OAuth 2.0 / OpenID Connect.
10.	Machine Learning Model	Not applicable to the core ACL decision; access is determined by explicit field-value rules.
11.	Infrastructure (Server / Cloud)	Application deployment on cloud or local server. Cloud: containerized application + managed database; Local: development server configuration.

Table-2: Application Characteristics:

S.No	Characteristics	Description / Technology
1	Open-Source Frameworks	Python, FastAPI, PostgreSQL and standard open-source web tooling used for the application.
2	Security	Script-controlled ACL, least-privilege access, authentication, authorization, input validation and secure API communication.

Table-2: Application Characteristics (Continuation)

S.No	Characteristics	Description / Technology
3	Scalable Architecture	A modular service structure separates request handling, ACL evaluation and data storage; technology: containerized services / scalable cloud deployment.
4	Availability	Application and database can use managed cloud services, health checks and redundant deployment where required.
5	Performance	Keep ACL evaluation lightweight, index frequently evaluated fields, minimize database round trips and cache stable policy configuration where appropriate.

References:

<https://c4model.com/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>